

Rachael Lin

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EDUCATION

UC Berkeley

Current

Berkeley, CA

August 2026 - May 2029

- **Relevant Coursework:** General Chemistry, The Structure and Interpretation of Computer Programs, Computational Biology, Linear Algebra and Differential Equations, Discrete Mathematics

College of San Mateo

Student

San Mateo, CA

June 2020 - December 2025

- **Relevant Coursework:** Python Programming, Elementary Chemistry, Web Design, Chinese I/II/III, World History

Stanford Online High School

Student

Stanford, CA

August 2022 - May 2026

- **Relevant Coursework:** AP Calculus BC, AP Biology, C++, AP Computer Science A, Multivariable Calculus, AP Physics C, AP Statistics, Linear Algebra, Advanced Topics in Biological Research, Data Science, Differential Equations, Analytical & Organic Chemistry, Real Analysis, Complex Analysis, Data Structures & Algorithms

EXPERIENCES

California Institute of Technology: Thomson Lab

Researcher

Pasadena, CA

June 2024 - May 2026

- Developed a differentiable reinforcement learning model to simulate viscoelastic biofluid behavior
- Analyzed single-cell RNA sequencing datasets of T cell exhaustion—identified differentially expressed genes, gene programs, and temporal cell-fate matrices across perturbations to validate novel T cell culturing technique
- Introduced lineage-anchored optimal transport, integrating lineage barcodes into entropic Sinkhorn optimization to improve single-cell fate inference—recovered rare cell populations w/ 21% increase in cosine similarity

Stanford University School of Medicine: Lanz Lab

Intern

Palo Alto, CA

June 2025 - August 2025

- Investigated the maturation of autoreactive antibodies in multiple sclerosis by introducing point mutations via PCR-based site-directed mutagenesis, expressing monoclonal antibodies, and quantifying binding to PLP peptides using ELISA assays
- Characterized mutations potentially underlying the transition from anti-EBV antibodies to autoreactive antibodies, providing evidence for a common motif underlying antibody cross-reactivity

Uber Work Experience Program

Intern

Online

October 2023 - November 2023

- Developed UI/UX designs to propose a safe and scalable riding experience for teens to product managers

ACTIVITIES

FIRST Tech Challenge — Team 5206, Stanford Online High School

Captain

Stanford, CA

September 2021 - June 2026

- Coordinated w/ global team of 25+ members to assemble & program robot for competition
- Mentored 10+ students in Fusion 360 software, design techniques, and version control
- Expanded outreach efforts by hosting workshops, speaker events, school hackathon, posting STEM series on Youtube, and maintaining website/social media

Pixel Journal — Stanford Online High School

Editor-in-Chief

Stanford, CA

September 2022 - June 2026

- Organized w/ 10+ remote writers/editors to produce engaging issues under tight deadlines
- Authored 20+ articles across 4 columns on a variety of topics: alumni interviews, school events, and more
- Used strong communication skills to connect global school community

San Mateo County Tobacco Education Coalition

Youth Advisor

San Mateo, CA

September 2022 - Present

- Managed 3 county PSA contests with ~100 submissions each to encourage awareness about risks of tobacco
- Led local cleanups, collecting 1k+ tobacco products in total
- Utilized public speaking, creative writing, and community-building skills to persuade local policy makers to ensure tobacco retailers place warnings next to products

Biology Olympiad Club — Stanford Online High School

Club Leader

San Mateo, CA

September 2024 - June 2026

- Developed weekly slides and practice questions to prepare students for competing in USA Biology Olympiad
- Hosted school's first Biology Bowl (trivia), with 45+ participants

VOLUNTEERING

Tutoring

Peer Tutor

Millbrae, CA

March 2023 - Present

- SAT Reading and Writing tutor on Schoolhouse, impacting ~20 learners across 9 countries
- Nominated by instructors at Stanford Online High School as a peer tutor for: Advanced Topics in Biological Research, C++: Techniques and Algorithms, AP Statistics, Textual Analysis and Argumentation, Modes of Writing and Argumentation, History and Philosophy of Science

TEST SCORES

- **SAT (11th grade 2024):** 1580 out of 1600
- **AP's:** AP Biology (5), AP Calculus BC (5), AP English Literature and Composition (5), AP English Language and Composition (4), AP Statistics (4), AP Computer Science (4), AP Physics C Mechanics (4)

AWARDS & HONORS

- National Scholastic Art and Writing Awards: National Gold Medal (2022), National Silver Medal (2025), 4x Regional Gold Key (2023-2026), 6x Regional Honorable Mention (2024-2026)
- Third place in Golden Gate Science Fair for Computer Science (2024)
- Dean's List at College of San Mateo (2021, 2022)
- President's Volunteer Service Gold Award (2023, 2024)
- Millbrae Educational Excellence Award (2022)
- USA Biology Olympiad Certificate of Merit (2025)
- FIRST Tech Challenge Qualifying Tournament: Connect Award (2024), Control Award (2023)

SKILLS

- **Languages and Tools:** Python (PyTorch, TensorFlow, NumPy, Scanpy, Matplotlib), C++, Figma, HTML, Java, R (Seurat), Fusion 360
- **Wet Lab:** Site-directed mutagenesis, PCR, plasmid purification, antibody expression/purification, ELISA